

ADITYA UNIVERSITY

AUS 25
STUDENTS MANUAL



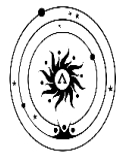
LABORATORY MANUAL

ENGINEERING GRAPHICS
(2501ME01)

Prepared by
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Verified by
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DEPARTMENT OF MECHANICAL ENGINEERING
2025-2026



ADITYA UNIVERSITY

Vision

- To be a globally recognized university through excellence in Education, Innovation and Sustainable growth.

Mission

- Deliver collaborative education to prepare students for global challenges through
- Transformative learning
- Vibrant research ecosystem
- Sustainable and inclusive community



ADITYA
UNIVERSITY

Department of

.....

.....

Certified that this is the bonafide record of
Practical work done by

Mr./Ms.

With Roll No. a student ofsemester

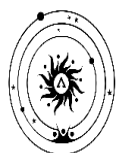
In the course during the Academic Year.....

No. of Experiments Conducted

No. of Experiments Completed

FACULTY

HEAD OF THE DEPARTMENT



ADITYA UNIVERSITY

1. Rubrics for Laboratory Course (L&T, P, C: 0,2,2 / 1,2,3)

Continuous Internal Evaluation (CIE) – 50 Marks		Semester End Examinations (SEE)- 50 Marks
Day-to-Day Evaluation	Internal Lab Examination	External Lab
25 Marks	25 Marks	50 marks

Day to Day – 25 Marks (Experiments performed on daily Basis & Record– 10 Marks, Regularity – 5 Marks and Viva- 10 Marks)			
Performance Indicator	Needs Improvement	Satisfactory	Good
P1: Daily performance of experiments and record submission (10 marks)	Could not complete the experimentation and submission of documents properly (Score: 1–3)	Partially completed the experimentation and/or submission of documents (Score: 4–7)	Completed the experimentation and submission of documents on time (Score: 8–10)
Evaluated at the end of semester			
P2: Participation & Regularity (5 marks)	Irregular in attending lab sessions; minimal participation in lab activities (Score: 1–2)	Attends most sessions; participates occasionally (Score: 3–4)	Regularly attends sessions and actively participates in lab work (Score: 5)
P3: Viva performance and analytical skills (10 marks) (Note: Average of two viva sessions conducted)	Unable to answer basic questions or lacks analytical thinking (Score: 1–3)	Answers some questions correctly; shows partial analytical ability (Score: 4–7)	Answers confidently and demonstrates strong analytical and problem-solving skills (Score: 8–10)

Student Copy to be maintained in the record:

DAY-TO-DAY EVALUATION

Roll No:

Academic Year:

Name of the Student:

Semester:

Name of Lab:

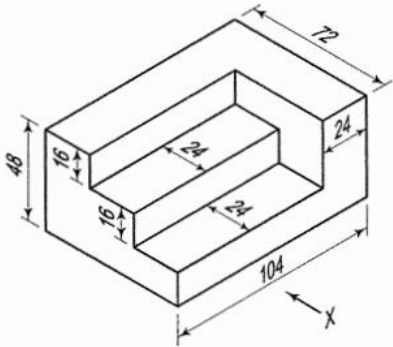
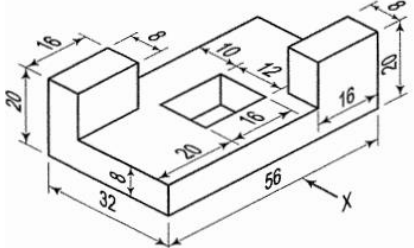
Course Code:

S.No.	NAME OF THE EXPERIMENT	P1: Experiments performed daily & submission of record (10 M)	Faculty sign
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
Total (P1)			
Average (P1) – 10 marks			

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Exp. No.		List of Experiments	
1		Introduction to AutoCAD	12
2	a	To draw a cycloid of given diameter of the generating circle is 40mm	
	b	To draw an Epicycloid, given the generating and directing circles of radii “r” is 25mm and “R” is 75mm.	
	c	To draw a Hypocycloid, given the generating and directing circles of radii “r” is 20mm and “R” is 65mm	
	d	To draw an Involute of given diameter of the generating circle is 40mm.	
	e	To draw an Involute of pentagonal plate of 30mm side.	
3	a	To draw the projections of the points in the following positions on the same ground line, keeping the projectors 25 mm apart. A, in the H.P. and 20 mm behind the V.P. B, 40 mm above the H.P. and 25 mm in front of the V.P. C, in the V.P. and 40 mm above the H.P. D, 25 mm below the H.P. and 25 mm behind the V.P. E, 15 mm above the H.P. and 50 mm behind the V.P. F, 40 mm below the H.P. and 25 mm in front of the V.P. G, in both the H.P. and the V.P.	
4	a	To draw the projections of the straight line in the below mentioned positions. 1. A 75mm long line parallel to both H.P and V.P and 25 mm from each. 2. A 75mm long line parallel 30 mm above H.P and in the V.P. 3. A 75mm long line parallel 40 mm in front of V.P and in the H.P.	

	b	To draw the projections of the following. 1. A 75 mm long line perpendicular to the H.P., 20 mm in front of the V.P. and its one end 15 mm above the H.P. 2. A 75 mm long line Perpendicular to the V.P., 25 mm above the H.P. and its one end in the V.P. 3. A 75 mm long line Perpendicular to the H.P., in the V.P. and its one end in the H.P.	
5	a	To draw a Line which is inclined at 30° to the H.P. and its one end 25 mm above it; parallel to and 25 mm in front of the V.P. Length of the Line is 75mm	
	b	To draw a Line which is inclined at 30° to the V.P. and its one end 20 mm above HP; parallel to and 20 mm in front of the V.P. Length of the Line is 75mm.	
	c	To draw the projections of a line AB 80 mm long has its end A 20 mm above the HP and 30 mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw the projections of the line.	
6	a	To draw a pentagonal plate of side 30mm is placed with one side on the HP and the surface inclined at 50° to HP and perpendicular to VP. Draw the projection of lamina.	
	b	To construct a hexagonal plane of side 30mm side has a corner at 20mm from VP and 50 mm from HP. Its surface is inclined at 45° to VP and perpendicular to HP. Draw the projection of lamina.	
7	a	To draw the projections of a pentagonal sheet of 25 mm side resting on VP having its surface inclined at 30° to VP. it's one side is parallel to VP and inclined at 45° to HP.	
	b	To draw the projections of a hexagon of side 30mm having one of its sides in HP and top view inclined at 65° to the VP and the surfaces inclined at 35° to the HP.	
8	a	To Draw the projections of a pentagonal pyramid, base 30 mm edge and axis 50 mm long, having its base on the H.P. and an edge of the base parallel to the V.P. Also draw its side view.	
	b	To Draw the projections of (i) a cylinder, base 40 mm diameter and axis 50 mm long, and (ii) a cone, base 40 mm diameter and axis 50 mm long, resting on the H.P. on their respective bases.	
	c	To Draw the projection of hexagonal prism, side of base 25mm and axis 50mm long resting on one of its rectangular faces on the H.P., with axis of prism is perpendicular to V.P., draw its projections.	
9	a	To construct a pentagonal prism of base side 30 mm and axis length 60 mm rests on H.P. on one of the base corners with the base edges containing it being equally inclined to H.P. The axis is inclined at 45° to HP and parallel to VP. Draw the projections of the prism	
	b	A cylinder having diameter 30 mm and axis height 40 mm is resting with point of its base circle on H.P. such that the axis making an angle of 30° to H.P and parallel to V.P. Draw projections of solid.	
	c	To draw the projections of hexagonal pyramid having a base with a 30 mm side and 60 mm long axis, has an corner of its base on H.P. Its axis is parallel to the V.P and inclined at 35° to the H.P.	

10	a	To develop the pentagonal prism of base 30mm and axis is 60mm	
	b	To develop the hexagonal prism of base 30mm and axis is 60mm	
11	a	To develop the hexagonal pyramid of base 25mm and axis is 55mm	
	b	To develop the pentagonal pyramid of base 25mm and axis is 55mm.	
12	a	To draw the orthographic projection of the given isometric view of the object. 	
	b	To draw the orthographic projection of the given isometric view of the object. 	

ENGINEERING GRAPHICS
(Common to all branches except Ag.E)

	L	T	P	C
Course Code: 2501ME01	1	0	2	3

Course Outcomes:

At the end of the course, student will be able to:

- CO1:** Apply the principles of engineering drawing to construct Engineering curves
- CO2:** Construct projections of points and lines.
- CO3:** Demonstrate visualization skills of projections of planes.
- CO4:** Demonstrate visualization skills of projections of solids and development of surfaces
- CO5:** Construct isometric and orthographic views of simple solids.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1		2			1	1		
CO2	3	2	1		2			1	1		
CO3	3	2	1		2			1	1		
CO4	3	2	1		2			1	1		
CO5	3	2	1		2			1	1		

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	3	
CO5	3	

Practice:

1. Introduction to Engineering Graphics

Introduction to AutoCAD, Dimensioning, elements of dimensioning, system of dimensioning, and Conventions in Drawing.

2. Construction of Cycloids and Involute

- a) Construction of Cycloid, Epicycloid and Hypocycloid
- b) Involute of a pentagon and circle

3. Introduction to Orthographic Projection

- a) Principles of Projection
- b) Orthographic Projection – Four Quadrants.
- c) First angle & Third angle Projection with examples, reference plane, importance of reference lines or Plane.
- d) Projections of a point situated in any one of the four quadrants.

4. Projection of straight lines-I:

- a) Projections of straight lines parallel to both reference planes.
- b) Projections of straight lines perpendicular to one reference plane and parallel to other reference plane
- c) Projections of straight line parallel to one plane & inclined to another plane

5. Projection of straight lines-II:

- a) Projections of straight line inclined to both reference planes

6. Projection of planes:

- a) Regular planes perpendicular to both reference planes, Parallel to one reference plane and inclined to the other reference plane
Ex: Rectangle, Pentagon, Hexagon and Rhombus

7. Projection of planes

- a) Projections of Planes inclined to both reference planes Ex: Rectangle, Pentagon, Hexagon and Rhombus.

8. Projection of solids

- a) Axis Perpendicular to H.P and Axis Perpendicular to V.P
Ex: Pentagonal and Hexagonal Prisms, Pyramids, Cylinder and Cone
- b) Axis Parallel to H.P and V.P
Pentagonal and Hexagonal Prisms, Pyramids, Cylinder and Cone

9. Projection of solids

- a) Projection of Solids with axis inclined to one reference plane and parallel to another plane
Ex: Pentagonal and Hexagonal Prisms, Pyramids, Cylinder and Cone

10. Development of Surfaces

- a) Development of Prisms and Cylinder simple cases

11. Development of Surfaces

- a) Development of Pyramids and Cone simple cases

12. Conversion of Isometric views to Orthographic views

- a) Practice figure - 1
- b) Practice figure - 2

Additional Practice:

1. Conversion of Isometric views to Orthographic views

- a) Practice figure - 3
- b) Practice figure - 4

2. Conversion of Orthographic views to Isometric views

- a) Practice figure - 1
- b) Practice figure - 2

Text Books:

1. Engineering Drawing, N. D. Bhatt, Charotar Publishing House, 54th edition, 2024, ISBN : 9789385039706.
2. Engineering Drawing and Graphics , Venugopal, New Age Publications, 2nd edition, 2019, ISBN: 9788122415452.

Reference Books:

1. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2021, ISBN: 978-9385983177.
2. Computer Aided Engineering Graphics, T. Jeyapoovan, Vikas Publishing house, New Delhi, 1st Edition, 2023. ISBN: 9789356743199

Web Links:

1. <https://nptel.ac.in/courses/112103019/>
2. <https://academy.autodesk.com/authenticated-home-user>
3. <https://www.sciencedirect.com/book/9780080108391/engineering-drawing-from-the-beginning>

Practice No-1

Introduction to AutoCAD

AutoCAD is a powerful computer-aided design (CAD) software used by professionals in industries such as architecture, engineering, construction, and manufacturing to create detailed 2D and 3D drawings. It allows users to design and draft plans, schematics, blueprints, and other types of technical drawings with precision and efficiency.

Key Features of AutoCAD:

1. **Drawing Tools:** AutoCAD provides a wide range of tools to create lines, circles, arcs, and other geometric shapes, which can be used for both 2D and 3D designs.
2. **Editing Tools:** It includes features to modify drawings, such as moving, scaling, rotating, and trimming objects.
3. **Layers:** AutoCAD allows users to organize different elements of a drawing into layers, making it easier to manage and control visibility.

SCREEN SETTINGS:

UNITS →

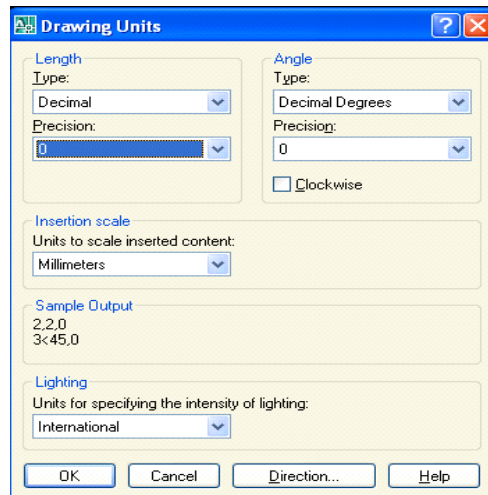
TYPE → Decimal

Precision → 0.00

Units scale → Millimeters

Units for specifying the intensity INTERNATIONAL

Click OK.



LIMITS →

0, 0 →

100, 100 →

GRID → 10 →

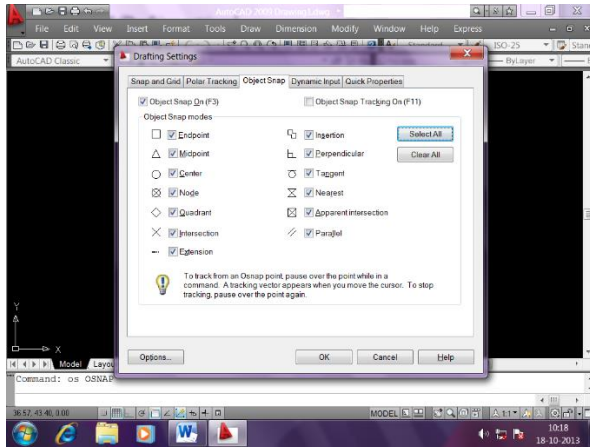
Z → (Zoom)

A → (All)

OS (Drafting settings)

OS →

Click Object Snap tab, click "Select all" snap modes option



D (Dimension Style Manager)

D →

Click Modify

Click Symbols & Arrows

Apply Arrow Size 3

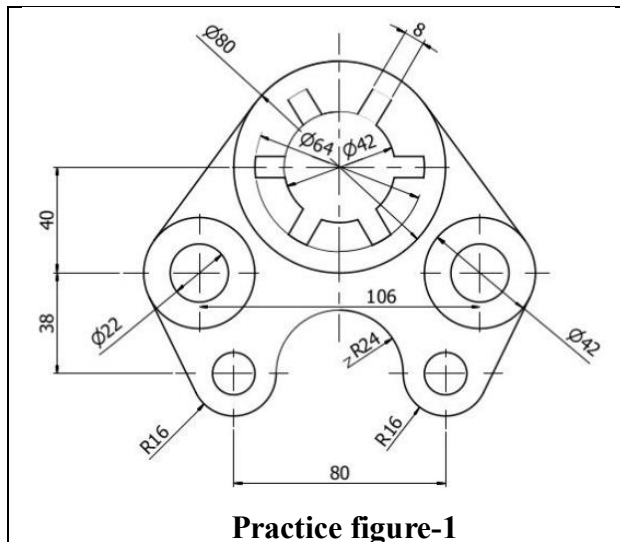
Click Text option

Apply Text height 4

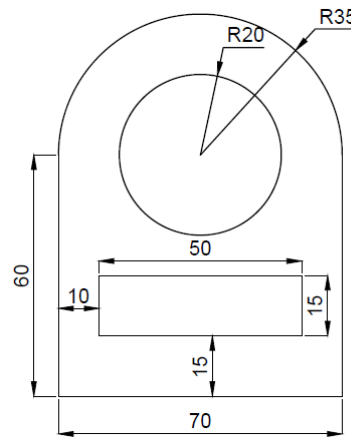
Commands Used in AutoCAD:

Draw Commands: Line (l), Polyline (pol), Circle (c), Arc (a), Rectangle (rec), Ellipse (el), Polygon (pol)

Modifying Commands: Move (m), Copy (co), Stretch (s), Rotate (ro), mirror (mi), scale (sc), Trim (tr), fillet (f), chamfer (ch), Array (ar)



Practice figure-1

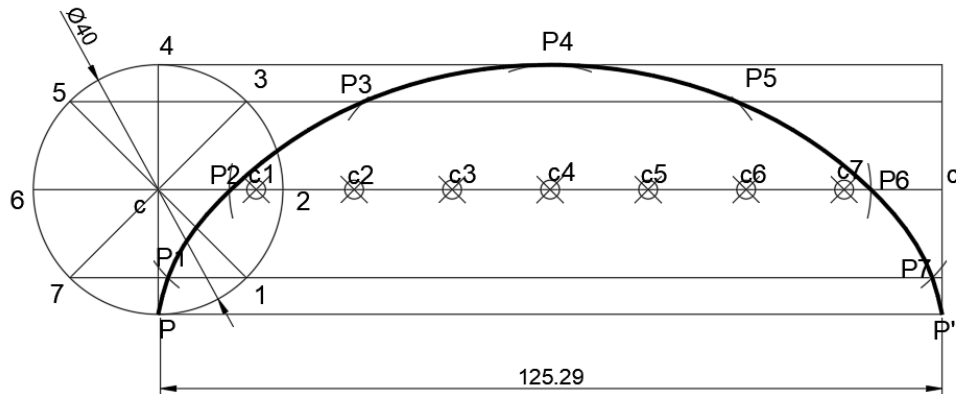


Practice figure-2

Practice No-2a

Aim: To draw a cycloid of given diameter of the generating circle is 40mm.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Type 'c' and press 'Enter' to activate the circle command.
2. Draw the circle with required diameter.
3. Divide the circle with 12 equal parts by using "array (polar array)" command.
4. Mark divisions 1 to 12 for circle by using 'MTEXT' command.
5. Activate the line command 'l' and draw the horizontal line from 'P' point on the circle with distance of circumference of the circle and mark the point Q.
6. Divide the line PQ into 12 equal parts by 'div' command.
7. Draw the horizontal lines passing through 11 to 01 and 10 to 02 etc. draw lines parallel to PA with line command 'l'.
8. Activate the arc command 'a' with centres C1, C2 etc. and radius equal to R, radius of generating circle, draw arcs cutting the lines through 1', 2' etc. at points P1, P2 etc. respectively. Activate the arc command 'a', draw a smooth curve through points P, P1, P2 A. This curve is the required cycloid.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.
3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The cycloidal curve generated with 40mm diameter of generating circle by using AutoCAD.

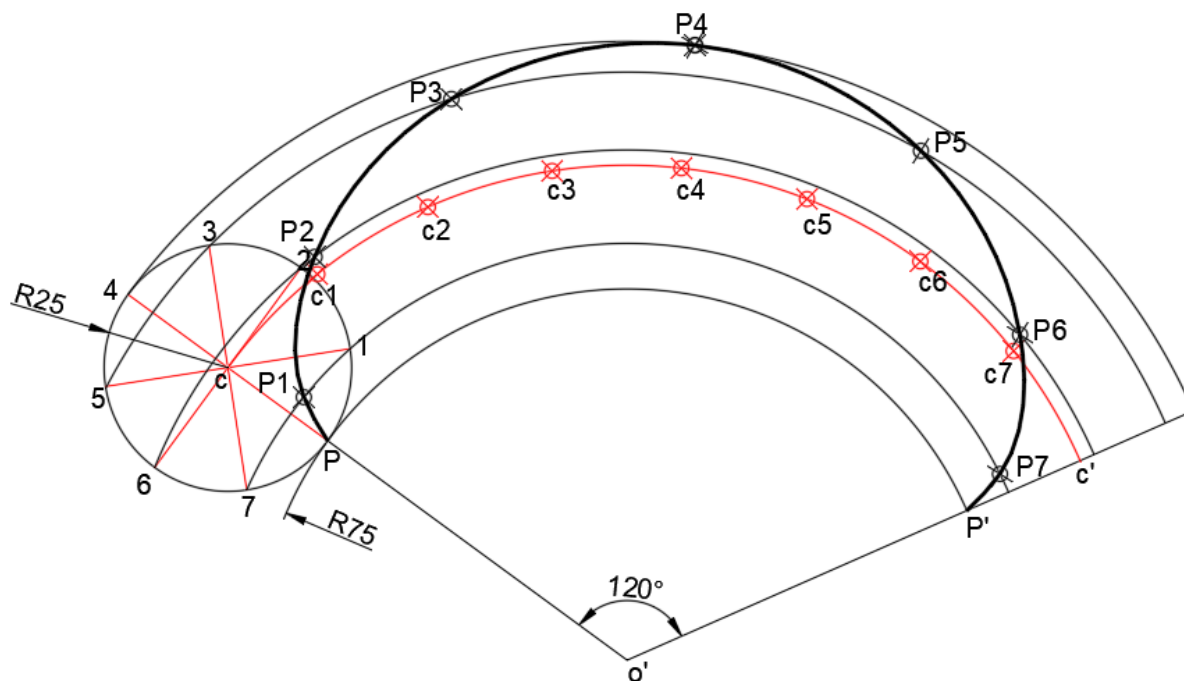
Viva Voce:

1. What is AutoCAD
2. How can you zoom in and out AutoCAD

Practice No-2b

Aim: To draw an Epicycloid, given the generating and directing circles of radii “r” is 25mm and “R” is 75mm.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Type 'c' and press 'Enter' to activate the circle command.
2. Draw the circle with required diameter.
3. Divide the circle with 12 equal parts by using “array (polar array)” command.
4. Mark divisions 1 to 12 for circle by using 'MTEXT' command.
5. Activate the line command 'l' and draw the line from 'P to O' with radius of directing circle.
6. The position of A may be located by calculating the angle subtended by the arc PA at centre O' by the formula, $\text{angle } PO'A = 360^\circ \times r/R$. Set-off this angle and obtain the position of A by activate line command 'l'.
7. With centre O' and radius equal to OO', draw an arc by using arc command 'a' intersecting O'A-produced at B. This arc OB is the locus of the centre O.
8. Divide OB with divide command 'div' and the generating circle into twelve equal parts.
9. With centre O', describe arcs through points 1' 2', 3' etc by divide command 'div'.
10. With centres O1, O2 etc. and radius equal to r, draw arcs by arc command 'a' cutting the arcs through 1 2 3 etc. at points P1, P2 etc.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.

3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Epicycloid curve generated with 50mm diameter of generating circle by using AutoCAD.

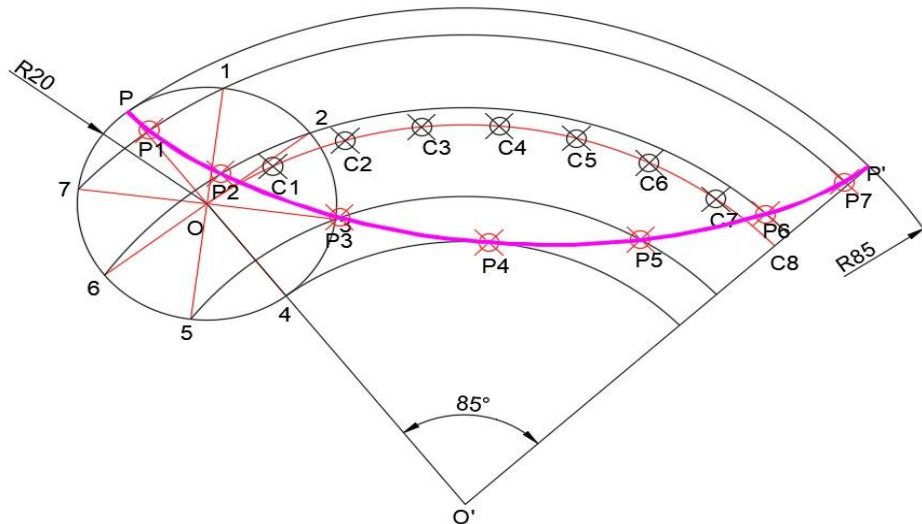
Viva Voce:

1. What is the function of OSNAP command in AutoCAD
2. How can you erase an object in AutoCAD

Practice No-2c

Aim: To draw a Hypocycloid, given the generating and directing circles of radii “r” is 20mm and “R” is 85mm.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Type 'c' and press 'Enter' to activate the circle command.
2. Draw the circle with required diameter.
3. Divide the circle with 12 equal parts by using “array (polar array)” command.
4. Mark divisions 1 to 12 for circle by using 'MTEXT' command.
5. Activate the line command 'l' and draw the line from O to O' with radius of directing circle.
6. The position of B may be located by calculating the angle subtended by the arc OB at centre O' by the formula, $\text{angle } OO'B = 360^\circ \times r/R$. Set-off this angle and obtain the position of B by activate line command 'l'.
7. With centre O' and radius equal to O'P, draw an arc by using arc command 'a' intersecting O'A.
8. Divide OB with divide command 'div' and the generating circle into twelve equal parts.
9. With centre O', describe arcs through points 1' 2', 3' etc by divide command 'div'.
10. With centres O1, O2 etc. and radius equal to r, draw arcs by arc command 'a' cutting the arcs through 1 2 3 etc. at points P1, P2 etc.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.
3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Hypocycloid curve generated with 40mm diameter of generating circle by using AutoCAD.

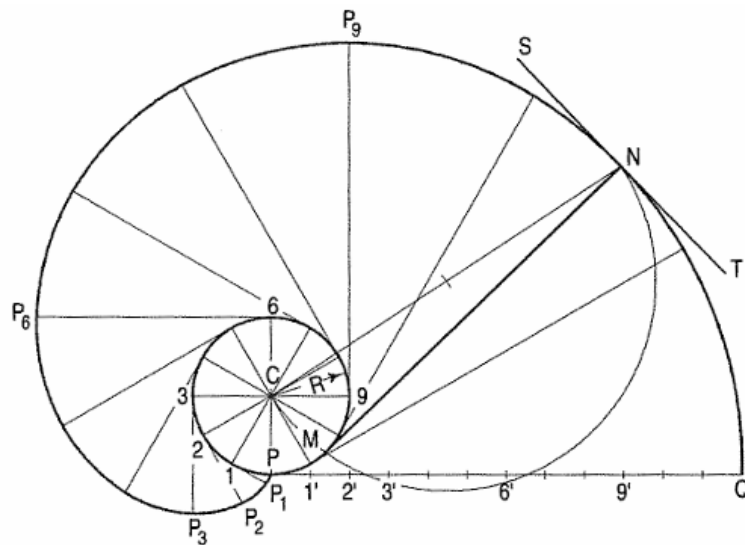
Viva Voce:

1. What is the function of OFFSET command in AutoCAD
2. List out the Draw Toolbar Commands in AutoCAD

Practice No-2d

Aim: To draw an Involute of given diameter of the generating circle is 40mm.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Type 'c' and press 'Enter' to activate the circle command.
2. Draw the circle with required diameter.
3. Divide the circle with 12 equal parts by using "array (polar array)" command.
4. Mark divisions 1 to 12 for circle by using 'MTEXT' command.
5. Activate the line command 'l' and draw the horizontal line from 'P' point on the circle with distance of circumference of the circle and mark the point Q.
6. Divide the line PQ into 12 equal parts by 'div' command.
7. Draw tangent line from 1, 2, 3 etc., by line command 'l' and mark on them points P1, P2, P3 etc., by arc command 'a' such that $1P1 = P1'$, $2P2 = P2'$, $3P3 = P3'$ etc.
8. Activate the arc command 'a' and draw the involute curve through the points P, P1, P2 ... etc.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.
3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Involute curve generated with 40mm diameter of generating circle by using AutoCAD.

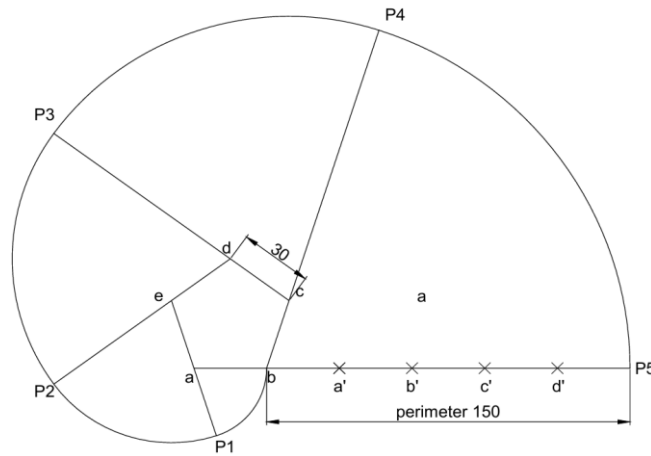
Viva Voce:

1. How can you draw the straight line in AutoCAD
2. List out some modify commands in AutoCAD

Practice No-2e

Aim: To draw an Involute of pentagonal plate of 30mm side.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Draw the pentagon by using polygon command 'pol', select 5 sides, type edge command 'e' press 'enter' and type 30mm.
2. Name the corners as a, b, c, d, and e by 'MT' Command.
3. Draw the perimeter line 150mm by line command 'l' from point 'b' on pentagon (5 sides * 30 = 150).
4. Mark divisions 1 to 12 for circle by using 'MTEXT' command.
5. Extend the 'ea' line by extend command 'ex' up to 30mm from point 'a' of pentagon similarly extend the 'de' line up to 60mm from point 'e', extend the 'dc' line up to 90mm from point 'd' etc.
6. Mark the points P1, P2, P3...
7. Draw the arc by select 'centre, start, end' option in AutoCAD and click on points 'abP1' to get the arc and similarly from same command draw the remaining arcs with points 'eP1P2' and 'dP2P3' etc.
8. Thus the curve obtained is involute.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.
3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Involute curve generated with side 30mm of pentagonal plate by using AutoCAD.

Viva Voce:

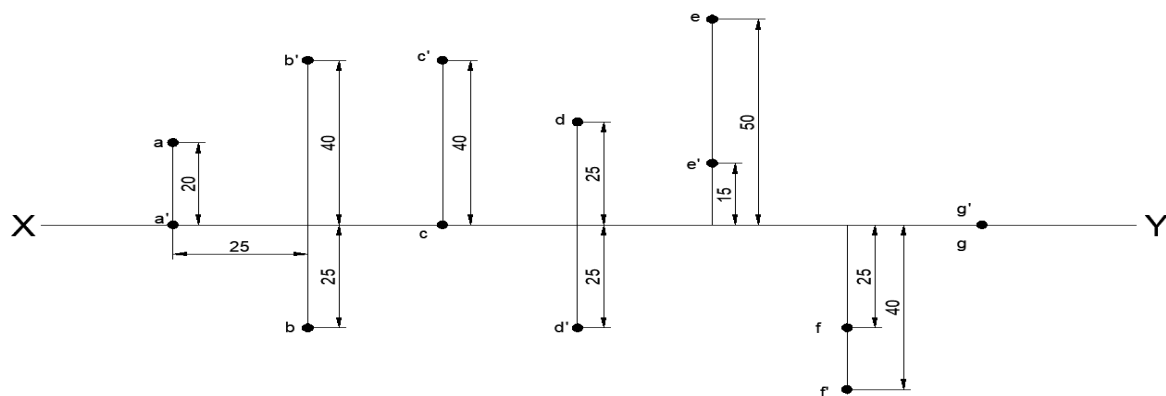
1. What is ARRAY command in AutoCAD
2. How many ways to draw the Rectangle in AutoCAD

Practice No-3

Aim: To draw the projections of the points in the following positions on the same ground line, keeping the projectors 25 mm apart.

A, in the H.P. and 20 mm behind the V.P.
B, 40 mm above the H.P. and 25 mm in front of the V.P.
C, in the V.P. and 40 mm above the H.P.
D, 25 mm below the H.P. and 25 mm behind the V.P.
E, 15 mm above the H.P. and 50 mm behind the V.P.
F, 40 mm below the H.P. and 25 mm in front of the V.P.
G, in both the H.P. and the V.P.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Type 'l' and press "Enter" to activate the line command.
2. Construct the line XY with the required length. (The dimension is not finite; it changes according to the space required for the solution. For the above case I took 195 mm)
3. Leave a distance of 25 mm from the projection lines of point 'A' and start representing the projections of point 'B'.
4. Repeat this procedure for all the other projections.
5. Active the donut command by typing 'do' and press 'Enter'. Give inside diameter as '0' and outside as '2'.
6. Specify the projections of point at the required locations using the activated donut command.

Precautions:

1. Make sure the limits are applied at the beginning.
2. Make sure the inside diameter of the donut is 'zero' and the outside is greater than 'zero' as per your need.

Result:

The projections of the given points are drawn on the same ground line keeping the projectors 25 mm apart.

Viva Voce:

1. How can you draw the circle in AutoCAD
2. What is the file extension of AutoCAD drawing files.

Practice No-4a

Aim: Draw the projections of the straight line in the below mentioned positions.

1. A 75mm long line parallel to both H.P and V.P and 25 mm from each.
2. A 75mm long line parallel 30 mm above H.P and in the V.P.
3. A 75mm long line parallel 40 mm in front of V.P and in the H.P.

Software required: AutoCAD

Procedure:

Case 1: Line Parallel to Both H.P. and V.P.

- Characteristics:
 - The line is parallel to both the Horizontal Plane (H.P.) and the Vertical Plane (V.P.).
 - Its projections on both planes will be straight lines parallel to the XY line.
- Projections:
 - Top View (Plan): The line's true length is visible in the top view, and it is parallel to the XY line.
 - Front View (Elevation): The line's true length is also visible in the front view, parallel to the XY line.
- Example:
 - If a line AB is 75 mm long and is 25 mm above H.P. and 25 mm in front of V.P., its projections will show AB as a 75 mm line parallel to the XY line in both the top and front views.

Case 2: Line in H.P. and Parallel to V.P.

- Characteristics:
 - The line lies in the Horizontal Plane (H.P.) and is parallel to the Vertical Plane (V.P.).
 - The entire length of the line is on the H.P., meaning its top view will be the line itself, and its front view will be a point.
- Projections:
 - Top View (Plan): The line appears as a straight line in its true length and is parallel to the XY line.
 - Front View (Elevation): Since the line lies entirely in the H.P., its front view will be a single point located on the XY line.
- Example:
 - If a line AB is 75 mm long and lies in 30mm above H.P with parallel and in the V.P., its top view will show AB as a 75 mm line parallel to the XY line, and the front view will show a point on the XY line.

Case 3: Line in V.P. and Parallel to H.P.

- Characteristics:
 - The line lies in the Vertical Plane (V.P.) and is parallel to the Horizontal Plane (H.P.).
 - The entire length of the line is on the V.P., meaning its front view will be the line itself, and its top view will be a point.
- Projections:
 - Front View (Elevation): The line appears as a straight line in its true length and is parallel to the XY line.
 - Top View (Plan): Since the line lies entirely in the V.P., its top view will be a single point located on the XY line.

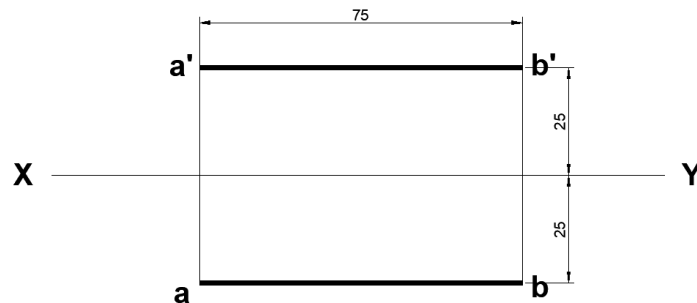
- Example:
 - If a line AB is 75 mm long and lies parallel to 40mm in front of V.P., parallel to the H.P., its front view 75mm in the H.P with parallel to the XY line.
 -

Summary of Line Projections

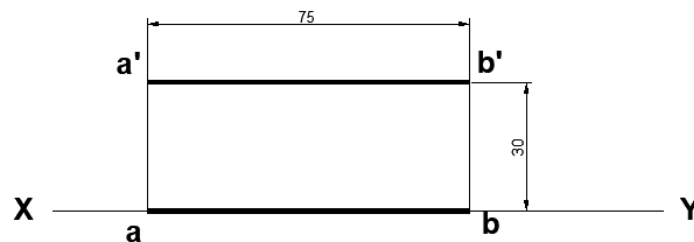
- Line Parallel to Both H.P. and V.P.:
 - Both front and top views will show the true length of the line, parallel to the XY line.
- Line in H.P. and Parallel to V.P.:
 - The top view will show the true length of the line, while the front view will be a point.
- Line in V.P. and Parallel to H.P.:
 - The front view will show the true length of the line, while the top view will be a point.

Solution:

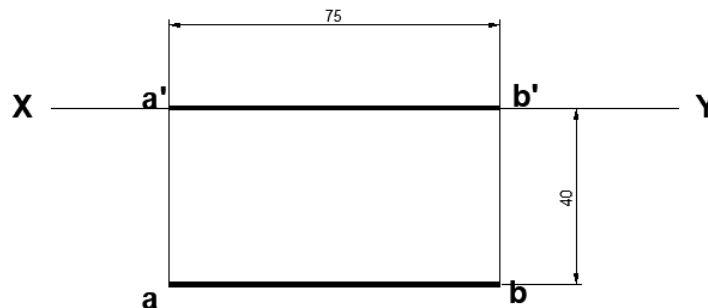
- A 75mm long line parallel to both H.P and V.P and 25 mm from each.



- A 75mm long line parallel 30 mm above H.P and in the V.P.



- A 75mm long line parallel 40 mm in front of V.P and in the H.P.



Precautions:

3. Make sure the limits are applied at the beginning.
4. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Carefully select the locations for dimensioning.

Result:

Projections of the straight line in the below-mentioned positions are drawn.

1. A 75mm long line parallel to both H.P and V.P and 25 mm from each.
2. A 75mm long line parallel 30 mm above H.P and in the V.P.

A 75mm long line parallel 40 mm in front of V.P and in the H.P.

Viva Voce:

1. What is the function of OSNAP command in AutoCAD
2. How can you erase an object in AutoCAD

Practice No-4b

Aim: To draw the projections of the following.

1. A 75 mm long line perpendicular to the H.P., 20 mm in front of the V.P. and its one end 15 mm above the H.P.
2. A 75 mm long line Perpendicular to the V.P., 25 mm above the H.P. and its one end in the V.P.
3. A 75 mm long line Perpendicular to the H.P., in the V.P. and its one end in the H.P.

Software Required: AutoCAD 2023

Procedure:

Line Perpendicular to H.P and Parallel to V.P

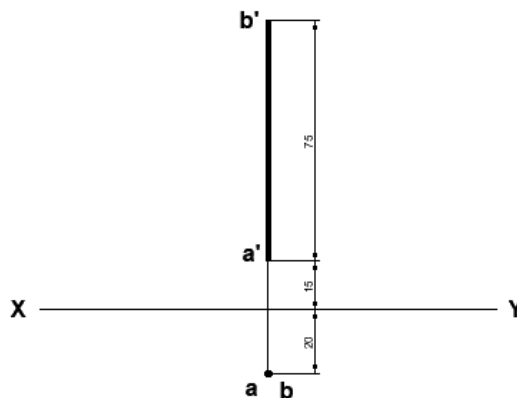
- Projection: When a line is perpendicular to the Horizontal Plane (H.P) and parallel to the Vertical Plane (V.P), its projection onto the H.P is a point.
- Reason: Since the line is perpendicular to H.P., it intersects H.P. at a single point. When projected onto H.P., this single point remains the same.
- Visual Example: Imagine a pole standing straight up in the ground. This pole is perpendicular to the ground (H.P) and parallel to the side of a building (V.P). If you look down from above (H.P. perspective), the pole appears as a single point.

Line Perpendicular to V.P and Parallel to H.P

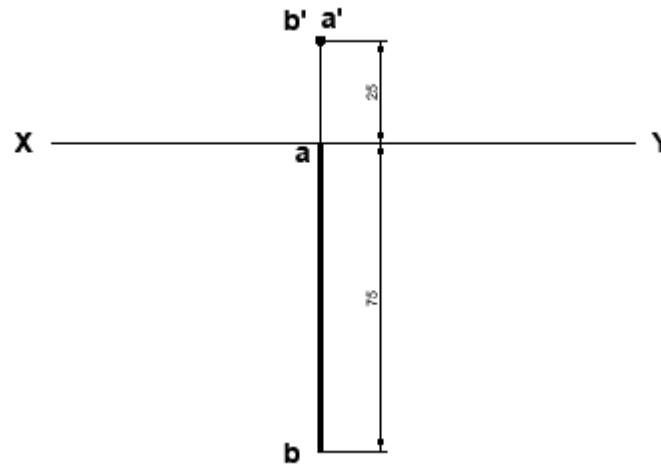
- Projection: When a line is perpendicular to the Vertical Plane (V.P) and parallel to the Horizontal Plane (H.P), its projection onto the V.P is a point.
- Reason: Since the line is perpendicular to V.P, it intersects V.P at a single point. When projected onto V.P, this single point remains the same.
- Visual Example: Imagine a pole lying flat on the ground (H.P). This pole is parallel to the ground and perpendicular to the side of a building (V.P). If you look at the building from the side (V.P perspective), the pole appears as a single point.

Solution:

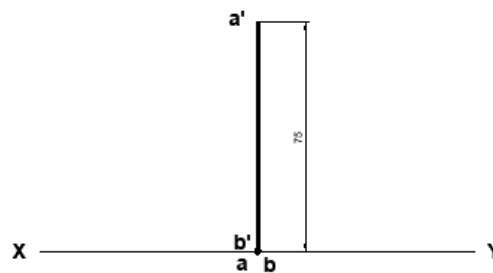
1. A 75 mm long line perpendicular to the H.P., 20 mm in front of the V.P. and its one end 15 mm above the H.P.



2. A 75 mm long line Perpendicular to the V.P., 25 mm above the H.P. and its one end in the V.P.



3. A 75 mm long line Perpendicular to the H.P., in the V.P. and its one end in the H.P.



Precautions:

1. Make sure the limits are applied at the beginning.
2. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.
3. Carefully select the locations for dimensioning.

Result:

projections of the following are drawn.

1. A 75 mm long line perpendicular to the H.P., 20 mm in front of the V.P. and its one end 15 mm above the H.P.
2. A 75 mm long line Perpendicular to the V.P., 25 mm above the H.P. and its one end in the V.P.
3. A 75 mm long line Perpendicular to the H.P., in the V.P. and its one end in the H.P.

Viva Voce:

1. What is the function of PEDIT command in AutoCAD
2. List out the draw commands in AutoCAD

Practice No-5a

Aim: Draw a Line which is inclined at 30° to the H.P. and its one end 25 mm above it; parallel to and 25 mm in front of the V.P. Length of the Line is 75mm.

Software Required: AutoCAD 2023

Procedure:

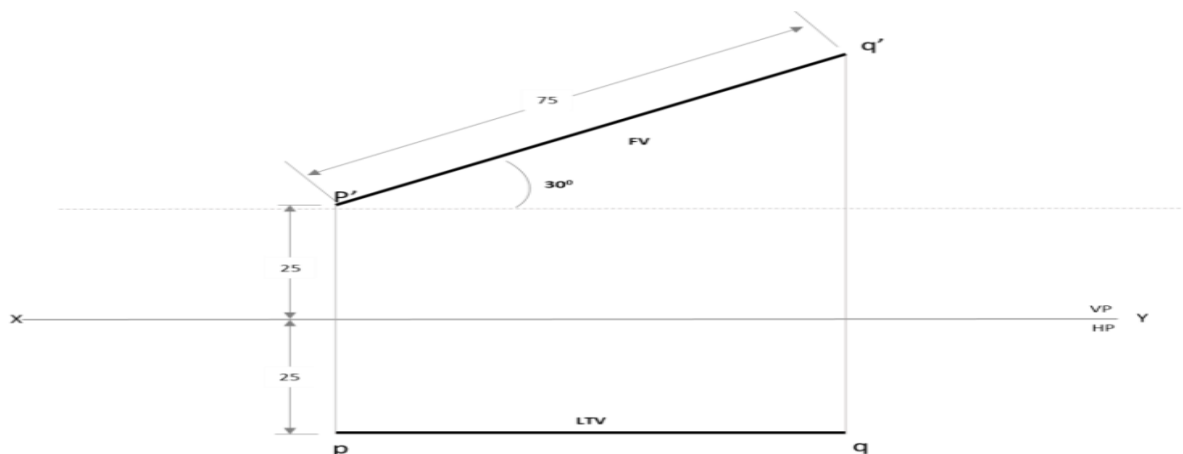
1- Type L at the command prompt and press the Enter key, or Click the Line icon tool from the Draw toolbar.

2- After you have selected the first point AutoCAD will ask you to specify the next point:

For Example

Enter @ 45, 0 to extend the line 45 units to the right from the last point, (Relative Cartesian coordinate entry).

Enter @50< 30° to extend the line 75 units 30° upward from the last point, (relative polar coordinate entry).



3. Draw XY line By Using Line Command L and Name with X, Y at two ends By Using Text Command

4. According to question 75 mm line 30° inclined and 25 mm above the H.P put the Point 25 mm from above XY line by Using Point Command Name the Point with p'

5. Line Parallel to and 25 mm in front of VP Put the point 25 mm below the XY line by using Point Command. Name the Point with p.

6. Draw 75 mm line from p' with inclination of 30° . Name the end point q' by using Text command Name it F.V

Result:

A line has been drawn, and the procedure for drawing a line inclined to the H.P. and parallel to the V.P., 25 mm from each, has been studied using AutoCAD.

Viva Voce:

1. What is the function of ORTHO mode in AutoCAD
2. What is the purpose of LAYERS in AutoCAD

Practice No-5b

Aim: Draw a Line which is inclined at 30° to the V.P. and its one end 20 mm above HP; parallel to and 20 mm in front of the V.P. Length of the Line is 75mm.

Software Required: AutoCAD 2023

Procedure:

1- Type L at the command prompt and press the Enter key, or Click the Line icon tool from the Draw toolbar.

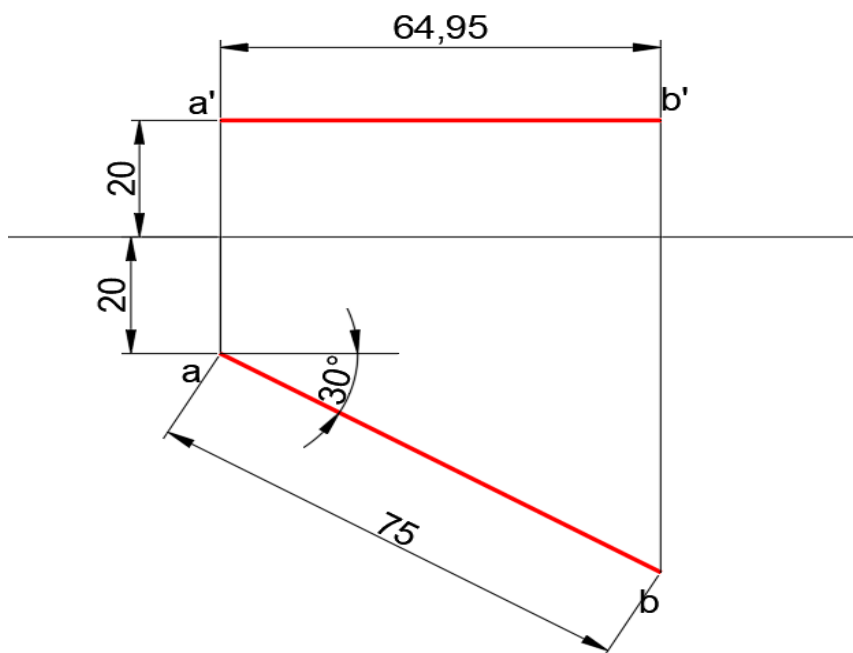
2- In any case, AutoCAD will prompt you with a message asking you to specify the first point to begin your line. - You can type the coordinates (x, y) of a point, and press enter to start the line, (Cartesian coordinate entry).

3- After you have selected the first point AutoCAD will ask you to specify the next point:

For Example

Enter @ 45, 0 to extend the line 45 units to the right from the last point, (Relative Cartesian coordinate entry).

Enter @50< 30° to extend the line 75 units 30° upward from the last point, (relative polar coordinate entry).



4. Draw XY line By Using Line Command L and Name with X, Y at two ends By Using Text Command

5. According to question 75 mm line 30° inclined and 20 mm above the H.P put the Point 20 mm from above XY line by Using Point Command Name the Point with p'

6. Line Parallel to and 20 mm in front of VP Put the point 20 mm below the XY line by using Point Command. Name the Point with p.

7. Draw 75 mm linen from p' with inclination of 30. Name the end point q' by using Text command Name it F.V

Result:

A line has been drawn, and the procedure for drawing a line inclined to V.P. and parallel to the H.P., 20 mm from each, has been studied using AutoCAD.

Viva Voce:

1. How can you draw Polygon in AutoCAD
2. What is the difference between COPY and MIRROR commands in AutoCAD

Practice No-5c

Aim: To draw the projections of a line AB 80 mm long has its end A 20 mm above the HP and 30 mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw the projections of the line.

Software Required: AutoCAD 2023

Procedure:

1- Type L at the command prompt and press the Enter key, or Click le Line icon tool from the Draw toolbar.

2- In any case, AutoCAD will prompt you with a message asking you to specify the first point to begin your line. - You can type the coordinates (x, y) of a point, and press enter to start the line, (Cartesian coordinate entry).

3- After you have selected the first point AutoCAD will ask you to specify the next point:

For Example

Enter @ 45, 0 to extend the line 45 units to the right from the last point, (Relative Cartesian coordinate entry).

Enter @50< 30° to extend the line 75 units 30° upward from the last point, (relative polar coordinate entry).

4. Draw XY line By Using Line Command L and Name with X, Y at two ends by using Text Command.

5. According to question 80 mm line 30° inclined and 20 mm above the H.P put the Point 20 mm from above XY line by Using Point Command Name the Point with p'.

6. Draw 80 mm line from p' with inclination of 30° Name the end point q₁' by using Text command.

7. According to question 80 mm line 45° inclined and 30 mm in front of V.P put the Point 30 mm from above XY line by Using Point Command Name the Point with p.

8. Draw 80 mm line from p with inclination of 45° Name the end point q₂ by using Text command.

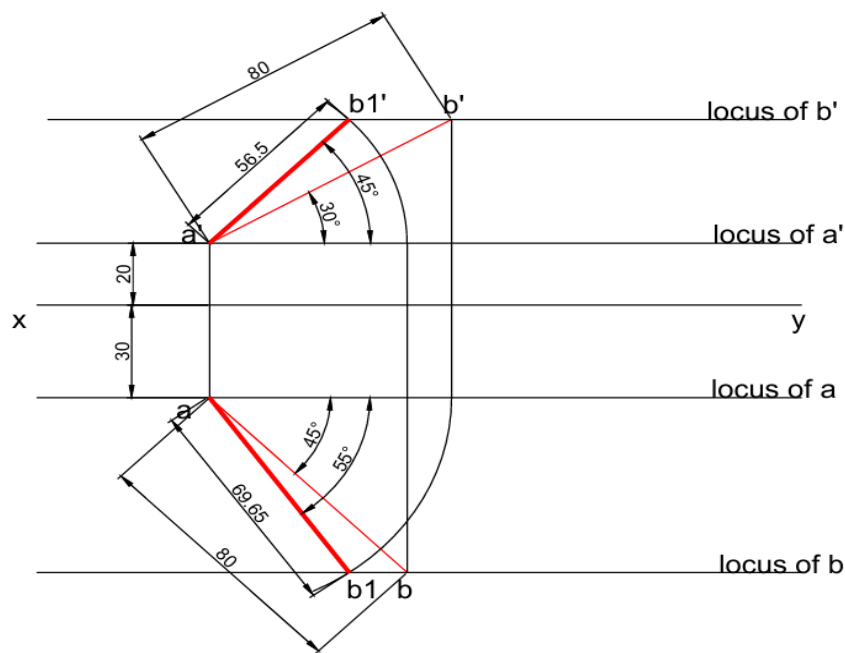
9. Draw perpendicular line from q₁' to locus of p name it as q₁. Draw another Perpendicular line from q₂ to locus of p₁ name the intersection point as q₂'.

10. Name p'q₂' Line as LFV and Name pq₁ line as Length of Top View.

11. For Final front View take P' as centre p'q₂' line as radius draw arc which will intersect Locus of P' at q' Join p'q' LINE it as FV.

12. For Final Top View take P as centre p'q₁' line as radius draw arc which will intersect Locus of P at q Join p q line it as TV

13. Mention the Dimensions by using DIMLINEAR Command.



Precautions:

1. Make sure the limits are applied at the beginning.
2. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.
3. Carefully select the locations for dimensioning.

Result:

A line has been drawn, and the procedure for drawing a line inclined to both the H.P. and V.P., has been studied using AutoCAD.

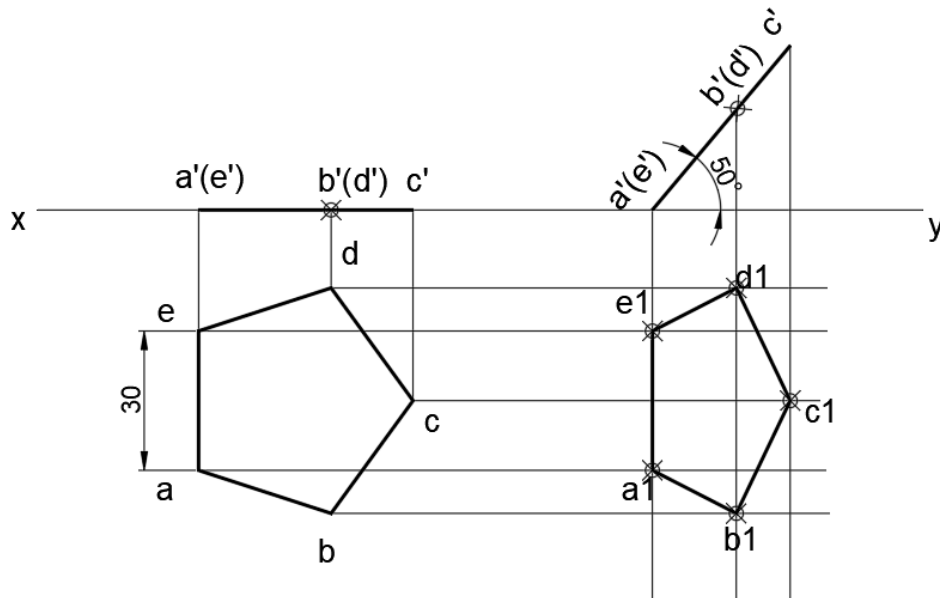
Viva Voce:

1. In AutoCAD 2D Modelling, which axis is not accessible for drafting? []
 A. X
 B. Y
 C. Z
 D. WCS
2. A Polyline can be broken into individual lines and arcs using which of the following command? []
 A. BREAK
 B. TRIM
 C. EXPLODE
 D. OVERKILL

Practice No-6a

Aim: To draw the pentagonal plate of side 30mm is placed with one side on the HP and the surface inclined at 50° to HP and perpendicular to VP. Draw the projection of lamina.

Software required: Auto CAD 2023



Solution: The plate makes an inclination with HP. Assume plate will be parallel to HP,
In order to get true shape of the plate will be visible in the TV.

Procedure:

1. Draw the reference line XY using line command.
2. Draw a pentagon a b c d e in the top view with one side is vertical.
3. Draw the top view appear like a line.
4. The front view is inclined with an angle 50° to H.P.
5. Draw the projection lines from front view, top view and intersection points and draw the top view after inclination of pentagon.
6. Save the figure in pdf format.

Precautions:

1. Set the drawing screen size to 297mm $\times$ 210mm. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Align the diagram must be fit to the A4 sheets.
4. Don't close the application until unless the job is plot successfully.

Result: The FV and TV of a given Square is drawn by using Auto CAD.

Viva Voce:

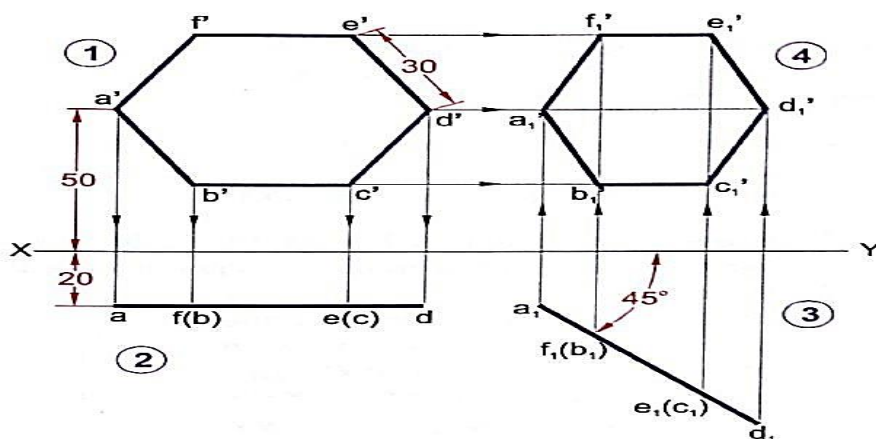
1. Origin of the AutoCAD drawing space is, []
 - A. 0,0
 - B. 1,0
 - C. 0,1
 - D. 1,1

2. Scrolling of mouse can perform which following action? []
 - A. Zoom in / zoom out
 - B. pan & scan
 - C. extents / all
 - D. scale

Practice No-6b

Aim: To draw a hexagonal plane of side 30mm side has a corner at 20mm from VP and 50 mm from HP. Its surface is inclined at 45° to VP and perpendicular to HP. Draw the projection of lamina.

Software required: Auto CAD 2023



Solution: The surface of the plane makes an inclination with VP. Assume plane will be parallel to VP, In order to get true shape of the plate it will be visible in the FV.

Procedure:

1. Draw the reference line XY using line command.
2. In Stage I draw hexagon of 30 mm in the FV using polygon command one of its corner is at a distance of 50mm above the reference line and give notation a', b', c', d', e', f' .
3. Project the FV keeping the line "a,d" parallel to and 20mm below the reference line xy.
4. Use copy command to copy the Stage I TV line (i.e., "a'd") and place it in the Stage II TV.
5. Rotate the copied TV with an angle of 45° using rotate command.
6. Project the Stage II TV and the Stage I FV. The intersection of these projection lines gives the final FV of a given figure.
7. Use "Dim" command to show all dimensions.
8. Save the figure in pdf format.

Precautions:

1. Set the drawing screen size to 297mm $\times$ 210mm. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Follow metric system such that all the dimension are in mm.
4. Align the diagram must be fit to the A4 sheets.
5. Don't close the application until unless the job is plotted successfully.

Result: The FV and TV of a given hexagonal plane is drawn by using Auto CAD.

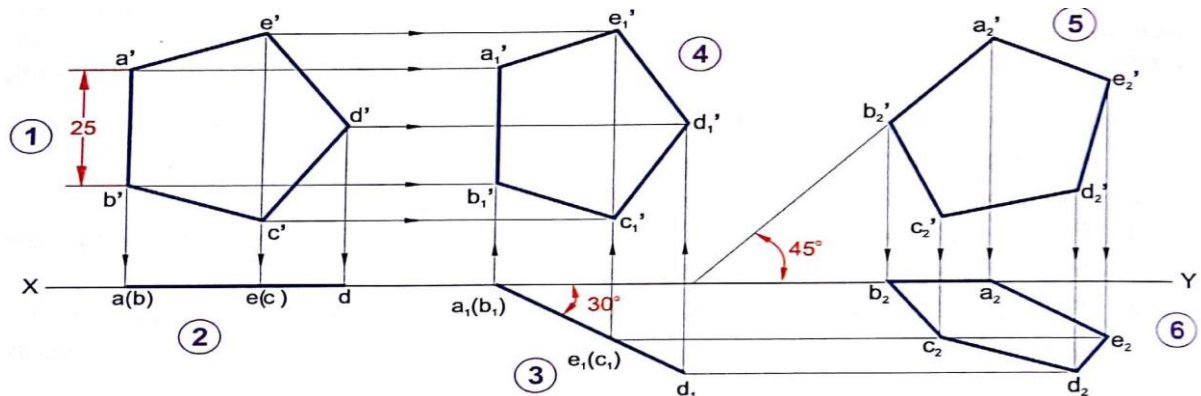
Viva Voce:

1. How can you draw Polygon in AutoCAD
2. What is the difference between COPY and MOVE commands in AutoCAD

Practice No-7a

Aim: To draw the projections of a pentagonal sheet of 25 mm side resting on VP having its surface inclined at 30° to VP. its one side is parallel to VP and inclined at 45° to HP.

Software required: Auto CAD 2023



Procedure:

1. Draw the reference line XY using line command.
2. In Stage I draw pentagon of 25 mm in the FV using polygon command with side condition and give notation a', b', c', d', e' .
3. Project the FV keeping the line "a,b,d" parallel to and its side "ab" is on the reference line xy.
4. Use copy command to copy the Stage I TV line (i.e., "a,b,d") and place it in the Stage II TV.
5. Rotate the copied TV with an angle of 30° using rotate command.
6. Project the Stage II TV and the Stage I FV. The intersection of these projection lines gives the FV with surface inclination.
7. Use copy command to copy the reduced shape of the pentagon from FV of the Stage II and place it in the FV region of Stage III.
8. Use Rotate command to Rotate the copied view so that the smaller side ab makes 45° to HP.
9. Project all the five points from the FV of Stage III and TV of Stage II.
10. Join the respective intersecting points of the projector lines to obtain the final projection point's a_2, b_2, c_2, d_2 and e_2 and obtain final TV of pentagonal shape.
11. Use "Dim" command to show all dimensions.
12. Save the figure in pdf format.

Precautions:

1. Set the drawing screen size to $297\text{mm} \times 210\text{mm}$. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Follow metric system such that all the dimension are in mm.
4. Align the diagram must be fit to the A4 sheets.
5. Don't close the application until unless the job is plotted successfully.

Result: The FV and TV of a given pentagonal plane is drawn by using Auto CAD

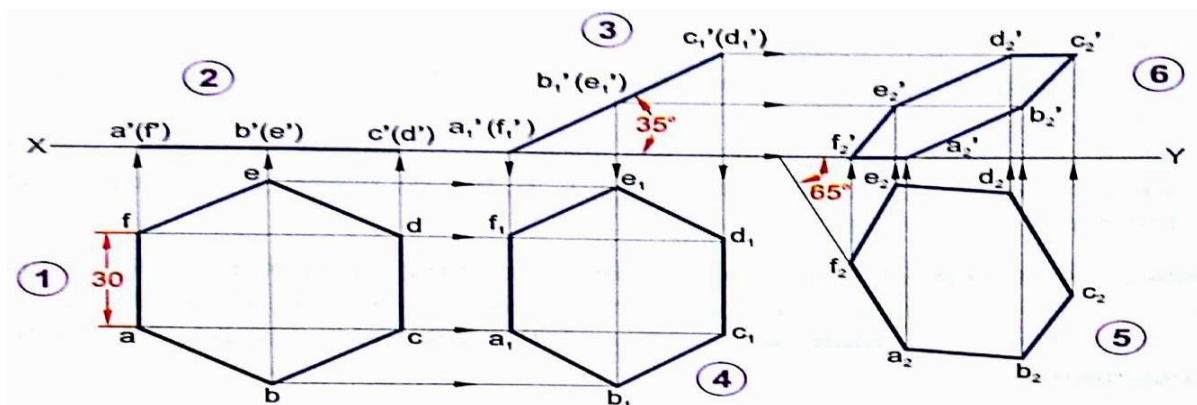
Viva Voce:

1. What is the shortcut for zoom command _____
2. What is the shortcut for polyline command _____

Practice No-7b

Aim: To draw the projections of a hexagon of side 30mm having one of its sides in HP and top view inclined at 65° to the VP and the surfaces inclined at 35° to the HP.

Software required: Auto CAD 2023



Procedure:

1. Draw the reference line XY using line command.
2. In Stage I draw hexagon of 30 mm in the TV using polygon command with side condition and give notation a,b,c,d,e,f.
3. Project the TV keeping the line "a'(f'),c'(d')'' parallel to and its side "a'(f')'' is on the reference line xy.
4. Use copy command to copy the Stage I FV line (i.e., "a₁(f₁),c₁(d₁)'' and place it in the Stage II FV.
5. Rotate the copied FV with an angle of 35° using rotate command.
6. Project the Stage II FV and the Stage I TV. The intersection of these projection lines gives the TV with surface inclination (reduced size hexagon).
7. Use copy command to copy the reduced shape of the hexagon from TV of the Stage II and place it in the TV region of Stage III.
8. Use Rotate command to Rotate the copied view so that the smaller side a₂,f₂ makes 65° to VP.
9. Project all the six points from the TV of Stage III and FV of Stage II.
10. Join the respective intersecting points of the projector lines to obtain the final projection point's a'₂, b'₂, c'₂, d'₂,e'₂ and f'₂ and obtain final hexagonal shape in the FV.
11. Use "Dim" command to show all dimensions.
12. Save the figure in pdf format.

Precautions:

1. Set the drawing screen size to 297mm $\times$ 210mm. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Follow metric system such that all the dimension is in mm.

Result: The FV and TV of a given hexagonal plane is drawn by using Auto CAD.

Viva Voce:

1. What is the function key for ORTHO mode _____
2. What is the function key for OSNAP _____

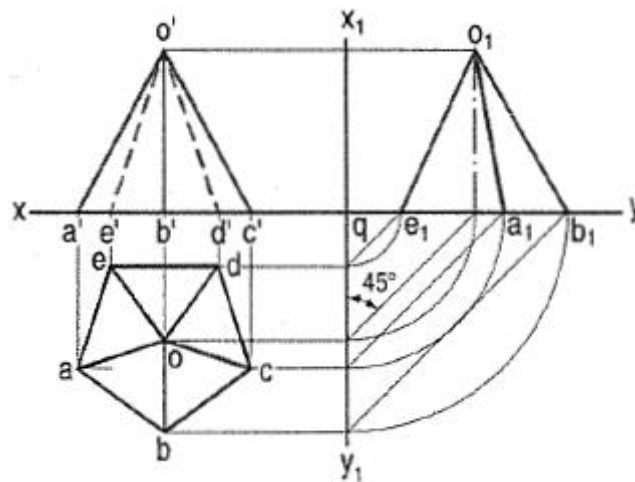
Practice No – 8a

Aim: To Draw the projections of a pentagonal pyramid, base 30 mm edge and axis 50 mm long, having its base on the H.P. and an edge of the base parallel to the V.P. Also draw its side view.

Software Required: AutoCAD 2023

Procedure:

1. Enter LIMITS 0,0 press enter and 210 ,297 press enter
2. Construct the axis line , xy with the required length.
3. Construct pentagon of side 30 mm in Topview.
4. Enter POL and press enter to execute the command, and when prompted to specify no. of sides, enter 5 and press enter. Specify the edge and move the cursor to specify the length of the edge.
5. Enter L and press enter to activate the line command draw the p rojections from a,b,c,e,d on to xy line to obtain a'e'b'd'c'.
6. Select line command and draw the axis length 50mm and draw the line by line command (L), and join o' to a'e'b'd'c' to form pyramid.
7. Draw the projection lines from a,b,c,d,e of pentagon and draw the arc from arc command (a), to obtain e1, a1, b1 and draw the axis line from side view with 50mm dimension and mark (o')
8. Using line command draw the side view of a pyramid.



Figure

Precautions:

- Always confirm that the drawing units are set to millimetres (or the required units) before starting. Use the UNITS command to check or set this.
- Use the TRIM, EXTEND, and OFFSET commands carefully to modify the drawing. Accidental trimming or extending can alter the geometry.

Result:

Hence by using AutoCAD Software we solved the FV and TV projections of pyramid.

Viva Voce:

1. What is the command for TRIM _____
2. What is the command for MULTITEXT _____

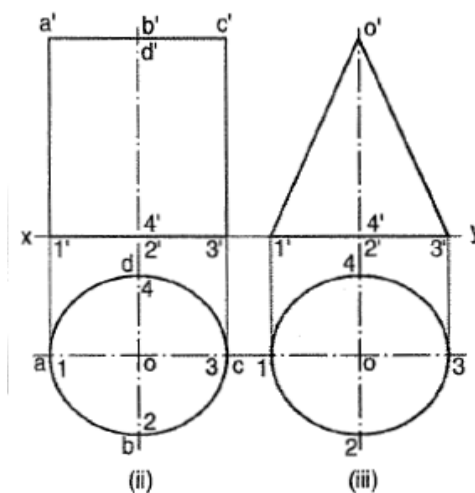
Practice No – 8b

Aim: To Draw the projections of (i) a cylinder, base 40 mm diameter and axis 50 mm long, and (ii) a cone, base 40 mm diameter and axis 50 mm long, resting on the H.P. on their respective bases.

Software Required: AutoCAD 2023

Procedure:

1. Enter LIMITS 0,0 press enter and 210 ,297 press enter
2. Construct the axis line , xy with the required length.
3. Construct two circles of diameter 40 mm in Topview.
4. Enter array command (AR) and divide the circles into 4 equal parts and give notations.
5. Enter L and press enter to activate the line command draw the projections from a1,b2,c3,d4 to xy line to obtain 1'2'3'4' for circle of cylinder and 1,2,3,4 to xy line to obtain 1'2'3'4' for circle of cone.
6. Select line command and draw the axis length 50mm for cylinder and cone draw the line by line command (L), and produce a'b'd'c'of cylinder and join (o') to 1'2'3'4' of cone.



Figure

Precautions:

- Always confirm that the drawing units are set to millimetres (or the required units) before starting. Use the UNITS command to check or set this.
- Use the TRIM, EXTEND, and OFFSET commands carefully to modify the drawing. Accidental trimming or extending can alter the geometry.

Result:

Hence by using AutoCAD Software we solved the FV and TV projections of cylinder and cone.

Viva Voce:

1. What is an object snap mode?
2. AutoCAD was first released in the year []
 - a) 1858
 - b) 1966
 - c) 1898
 - d) 1982

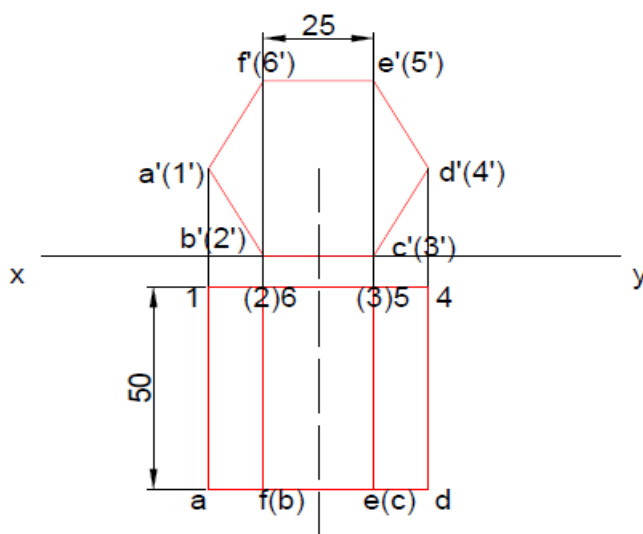
Practice No – 8c

Aim: To Draw the projection of hexagonal prism, side of base 25mm and axis 50mm long resting on one of its rectangular faces on the H.P., with axis of prism is perpendicular to V.P., draw its projections.

Software Required: AutoCAD 2023

Procedure:

1. Enter LIMITS 0,0 press enter and 210 ,297 press enter
2. Construct the axis line , xy with the required length.
3. Construct hexagon of side 25mm by using ‘pol’ command in Front view.
4. Mark the notations a’1’,b’2’,c’3’,d’4’,e’5’,f’6’.
5. Enter L and press enter to activate the line command draw the projection lines from a’1’,d’4’,e’5’,f’6’ on to xy line to obtain 1,2,3,4,5,6.
6. Select line command and draw the axis length 50mm and draw the line by line command (L), and to form topview of prism a,b,c,d,e,f.



Figure

Precautions:

- Always confirm that the drawing units are set to millimetres (or the required units) before starting. Use the UNITS command to check or set this.
- Use the TRIM, EXTEND, and OFFSET commands carefully to modify the drawing. Accidental trimming or extending can alter the geometry.

Result:

Hence by using AutoCAD Software we solved the FV and TV projections of pyramid.

Viva Voce:

1. Which mode allows the user to draw 90° straight lines []
 - a) Osnap
 - b) Ortho
 - c) Linear
 - d) Polar tracking

2. To obtain parallel lines, concentric circles and parallel curves; _____ is used.
 - a) Array
 - b) Fillet
 - c) Copy
 - d) Offset

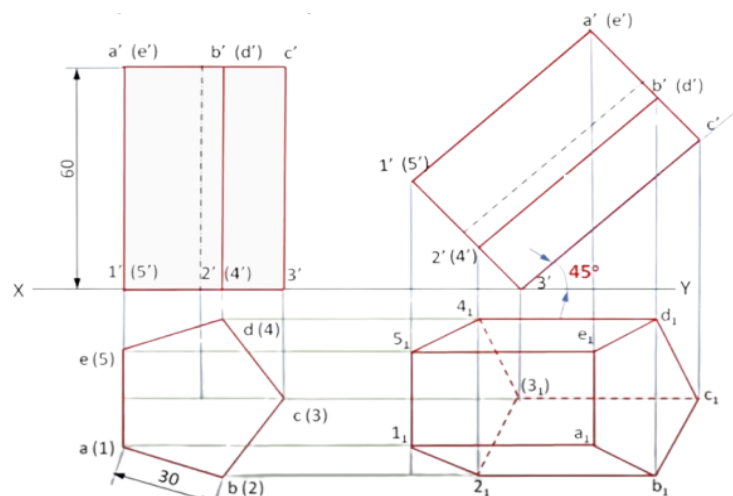
Practice No – 9a

Aim: To draw the pentagonal prism of base side 30 mm and axis length 60 mm rests on H.P. on one of the base corners with the base edges containing it being equally inclined to H.P. The axis is inclined at 45° to HP and parallel to VP. Draw the projections of the prism.

Software Required: AutoCAD 2023

Procedure:

1. Enter LIMITS 0,0 press enter and 210 ,297 press enter
7. Construct the axis line , xy with the required length.
8. Construct pentagon of side 30 mm in Topview.
9. Enter POL and press enter to execute the command, and when prompted to specify no. of sides, enter 5 and press enter. Specify the edge and move the cursor to specify the length of the edge.
10. Enter L and press enter to activate the line command draw the p rojections from e,d,c on to xy line to obtain 1'2',5'4'and 3.
11. Select line command and draw the axis length 60mm from center of the prism on xy line. Draw the other projections from 1'5' 3' to obtain first front view
12. Select the copy command and rotate command to tilt the front view by 45° to obtain second front view.
13. Using line command draw the projections of second top view on to the xy line.



Figure

Precautions:

- Always confirm that the drawing units are set to millimetres (or the required units) before starting. Use the UNITS command to check or set this.
- Use the TRIM, EXTEND, and OFFSET commands carefully to modify the drawing. Accidental trimming or extending can alter the geometry.

Result:

Hence by using AutoCAD Software we solved the FV and TV projections of prism.

Viva Voce:

1. The default grid spacing in both X and Y directions is []
 - a) 10
 - b) 20
 - c) 5
 - d) 15

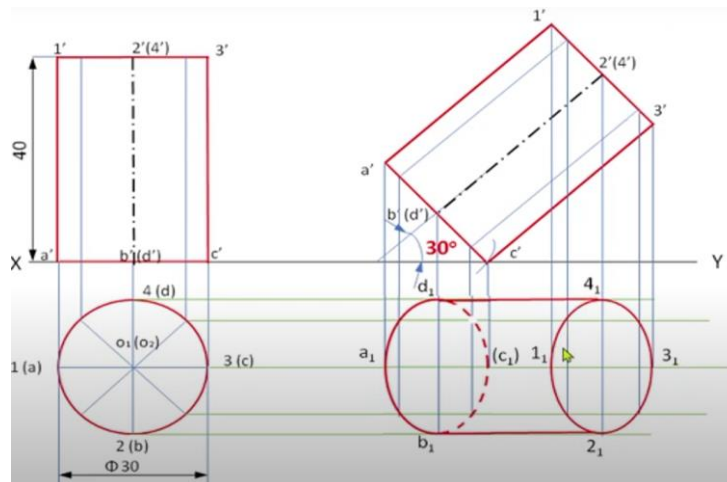
2. How many workspaces are available in AutoCAD []
 - a) 2
 - b) 4
 - c) 3
 - d) 5

Practice No-9b

Aim: A cylinder having diameter 30 mm and axis height 40 mm is resting with point of its base circle on H.P. such that the axis making an angle of 30° to H.P and parallel to V.P. Draw projections of solid.

Software required : AUTOCAD 2023

Solution drawn in AutoCAD:



Procedure:

1. Construct the axis line , xy with the required length.
2. Construct circle of diameter 30 mm in Topview.
3. Divide the circle into array command and give notations.
4. Project the projection lines and draw the front view with height 40mm.
5. Rotate the front view with rotate command and give 30 degrees inclination.
6. Project the front view of rotate object and top view of circle with projection lines.
7. Draw the top view with coincide points a_1, b_1, c_1, d_1 and $1_1, 2_1, 3_1, 4_1$.

Precautions:

1. Set Units and Scale Correctly.
2. Configure Drawing Limits.
3. Verify coordinates and measurements regularly to avoid inaccuracies.

Result:

A cylinder it's with the axis inclined to H.P and parallel to V.P with the given dimensions was successfully drafted using AUTOCAD software.

Viva Voce:

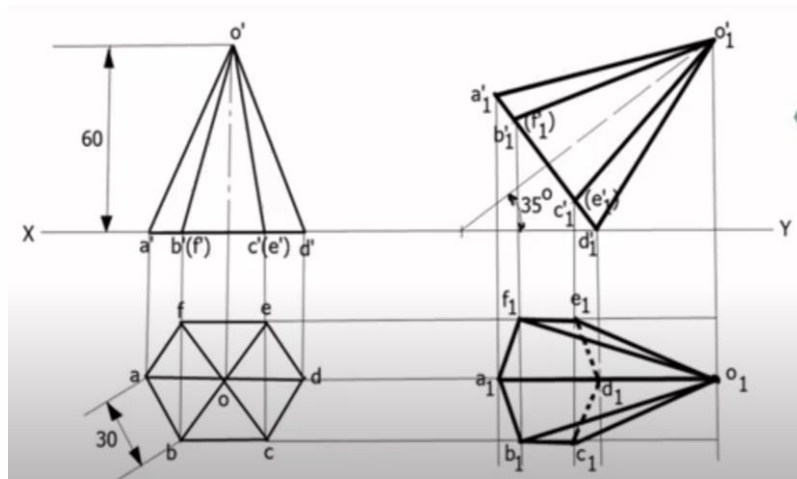
1. How many grip points does a circle have []
 - a) 5
 - b) 4
 - c) 3
 - d) 2

2. Which command is used to divide the object into segments []
 - a) Divide
 - b) Chamfer
 - c) Trim
 - d) Measure

Practice No-9c

Aim: To draw the projections of hexagonal pyramid having a base with a 30 mm side and 60 mm long axis, has an corner of its base on H.P. Its axis is parallel to the V.P and inclined at 35° to the H.P.

Software required : AutoCAD 2023



Solution

Procedure:

1. Enter LIMITS 0,0 press enter and 420 ,297 press enter
2. Construct the axis line , xy with the required length.
3. Construct hexagon of side 30 mm in Topview by using POL command.
4. Enter L and press enter to activate the line command(reduce the thickness of the line) and draw the projections from a,b,c,d,e,f on to xy line to obtain a'b',f'c' and e'd'
5. Select line command and draw the axis length 60 mm from center of the pyramid on xy line. Draw the other projections from a'b', e'd' to obtain first front view
6. Select the copy command and rotate command to tilt the front view by 35° to obtain second front view.
7. Using line command draw the projections of second top view on to the xy line.
8. Match all the points from second front view with first top view, join all the points by line command. Join all the edges and hidden lines to obtain the required pyramid a1,b1,c1,d1,e1,f1 and O1.

Precautions:

1. Make sure the blimits are applied at the beginning.
2. Use z enter and e enter to keep all the geometry inside the layout.
3. Carefully select the locationbns for dimensioning

Result:

The hexagonal pyramid with the axis inclined to H.P and parallel to V.P with the given dimensions was successfully drafted using AUTOCAD software.

Viva Voce:

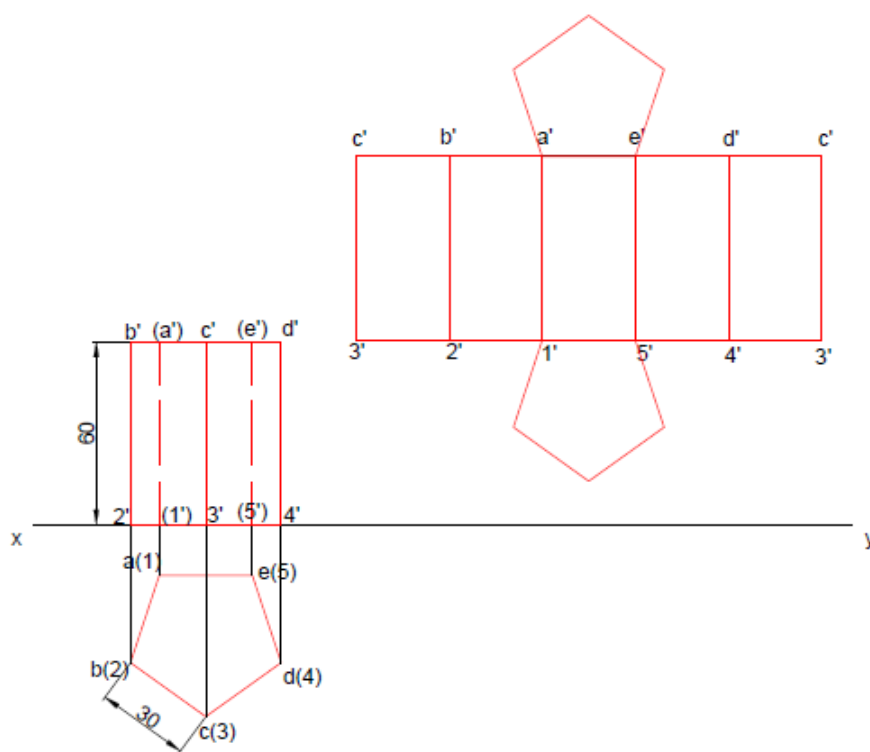
1. CAD/CAM is the relationship between []
 - a) science and engineering
 - b) manufacturing and marketing
 - c) design and manufacturing
 - d) design and marketing

2. Which axis is not used when working in 2-D frame []
 - a) Z-axis
 - b) Y-axis
 - c) X-axis
 - d) WCS

Practice No-10a

Aim: To develop the Pentagonal prism of base 30mm and axis is 60mm.

Software Required: AutoCAD 2023



Solution

Procedure:

1. Take the line command 'l' to draw a reference line with 'xy'
2. Draw the TV, pentagon by using polygon command 'pol' with 30mm side.
3. Draw the FV of prism by using line command 'l' with 60mm height.
4. Mark the notations by using 'mtext' command.
5. Draw the rectangle by using 'rec' command 'l' with 60mm height and 150mm length.
6. Take offset command divide the rectangle into 5 equal parts.
7. Draw the two pentagonal planes above and below by using 'pol' command.
8. Mark the notations by using 'mtext' command.

Precautions:

1. Make sure the units are selected in 'mm' at the beginning.
2. Make sure the limits are applied at the beginning.
3. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Development of pentagonal prism with 30mm side and 60mm height is drawn by using AutoCAD.

Viva Voce:

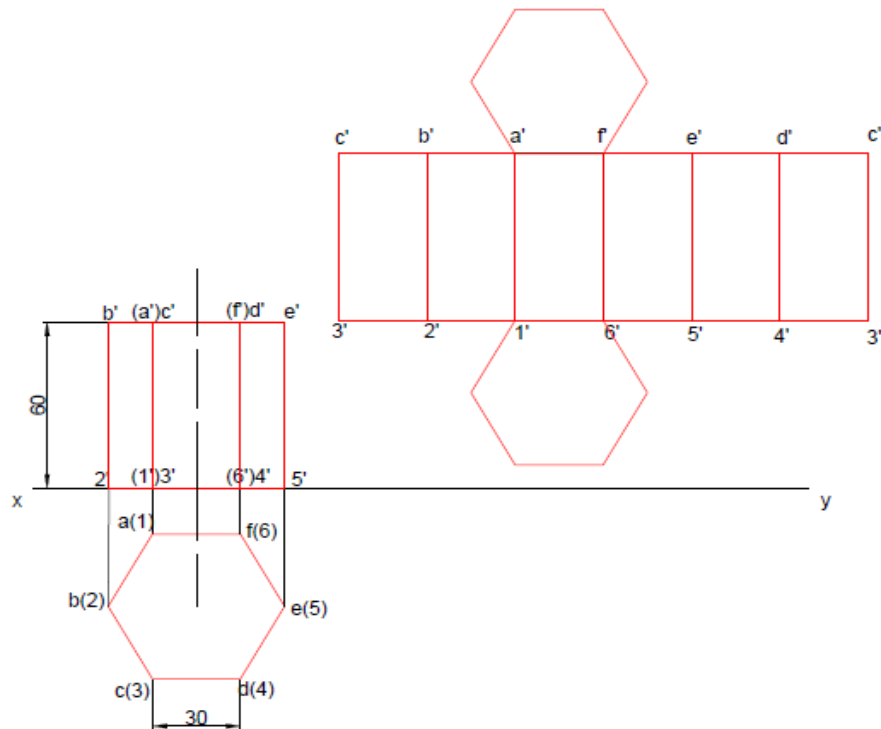
1. Which is corresponded to zoom mouse wheel? []
 - a) Extend/all
 - b) Zoom in/ Zoom out
 - c) Pan and scan
 - d) Scale

2. The offset command cannot be used to create _____
 - a) Concentric circles
 - b) Vertical straights
 - c) Three parallel lines
 - d) Parallel arcs

Practice No-10b

Aim: To develop the Hexagonal prism of base 30mm and axis is 60mm.

Software Required: AutoCAD 2023



Solution

Procedure:

9. Take the line command 'l' to draw a reference line with 'xy'
10. Draw the TV, hexagon by using polygon command 'pol' with 30mm side.
11. Draw the FV of prism by using line command 'l' with 60mm height.
12. Mark the notations by using 'mtext' command.
13. Draw the rectangle by using 'rec' command 'l' with 60mm height and 180mm length.
14. Take offset command divide the rectangle into 6 equal parts.
15. Draw the two hexagonal planes above and below by using 'pol' command.
16. Mark the notations by using 'mtext' command.

Precautions:

4. Make sure the units are selected in 'mm' at the beginning.
5. Make sure the limits are applied at the beginning.
6. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Development of hexagonal prism with 30mm side and 60mm height is drawn by using AutoCAD.

Viva Voce:

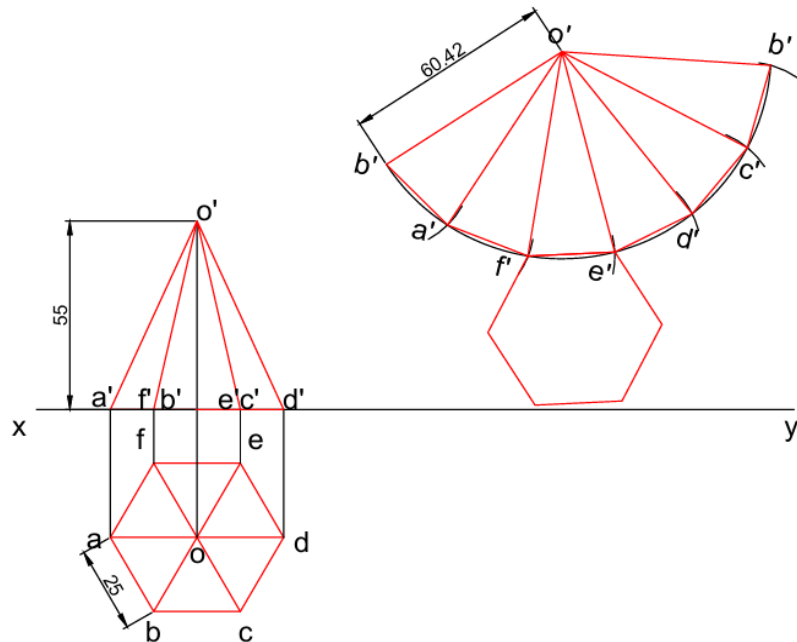
1. Function key F9 is used to toggle _____
 - a) snap
 - b) ortho
 - c) osnap
 - d) grid

2. Which function key is used to set Grid on or off? []
 - a) F7
 - b) F4
 - c) F3
 - d) F5

Practice No-11a

Aim: To develop the hexagonal pyramid of base 25mm and axis is 55mm.

Software Required: AutoCAD 2023



Solution

Procedure:

17. Take the line command 'l' to draw a reference line with 'xy'
18. Draw the TV, hexagon by using polygon command 'pol' with 25mm side.
19. Draw the FV of pyramid by using line command 'l' with 55mm height.
20. Mark the notations by using 'mtext' command.
21. Draw the slant height by using line command 'l' with any angle.
22. Draw the arc by arc command 'a' with slant height value.
23. Draw the 6 sides of hexagon cut by the arc command and line command 'l'.
24. Mark the notations by using 'mtext' command.

Precautions:

7. Make sure the units are selected in 'mm' at the beginning.
8. Make sure the limits are applied at the beginning.
9. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Development of hexagonal pyramid with 25mm side and 55mm height is drawn by using AutoCAD.

Viva Voce:

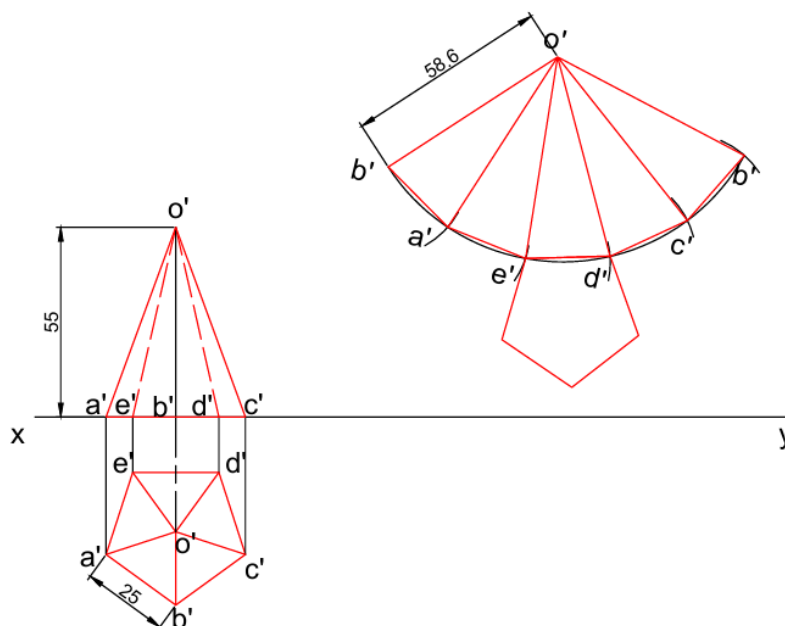
1. Symbol for quadrant snap, which appears on the entity while selecting is _____
 - a) ☐
 - b) ☐
 - c) ☐
 - d) ☐

2. Symbol for end point, which appears on the entity while selecting is _____
 - a) ☐
 - b) ☐
 - c) ☐
 - d) ☐

Practice No-11b

Aim: To develop the pentagonal pyramid of base 25mm and axis is 55mm.

Software Required: AutoCAD 2023



Solution

Procedure:

25. Take the line command 'l' to draw a reference line with 'xy'
26. Draw the TV, pentagon by using polygon command 'pol' with 25mm side.
27. Draw the FV of pyramid by using line command 'l' with 55mm height.
28. Mark the notations by using 'mtext' command.
29. Draw the slant height by using line command 'l' with any angle.
30. Draw the arc by arc command 'a' with slant height value.
31. Draw the 5 sides of hexagon cut by the arc command and line command 'l'.
32. Mark the notations by using 'mtext' command.

Precautions:

10. Make sure the units are selected in 'mm' at the beginning.
11. Make sure the limits are applied at the beginning.
12. If any of the pictorial information is going outside the boundary. Please use 'z' 'Enter' and 'E' 'Enter' to keep all the geometry inside the layout.

Result:

The Development of pentagonal pyramid with 25mm side and 55mm height is drawn by using AutoCAD.

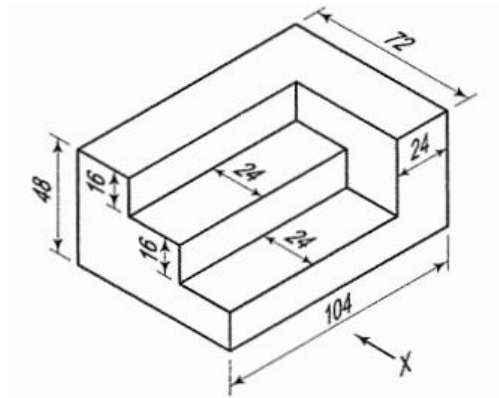
Viva Voce:

1. It is difficult to pick a point exactly and quickly with the mouse. In order to allow specifying a point exactly, which of the following command will help? []
 - a) Grid
 - b) Ortho
 - c) Snap
 - d) OSNAP

2. Object snap increases _____
 - a) speed, accuracy and ease of drafting
 - b) speed of drawing
 - c) memory utilization
 - d) accuracy while drafting

Practice No- 12a

Aim: To draw the orthographic projection of the given isometric view of the object.

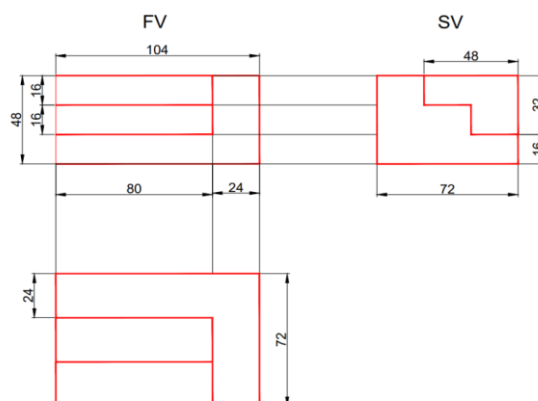


Software Required: AutoCAD 2023

Procedure:

1. Make necessary steps TV like units, limits and zoom. Pick the ortho mode on.
2. Access the Snap and Grid tab of the Drafting Settings dialog box
3. Observe the given isometric view. The arrow mark indicates the FV direction.
4. In the front view observe dimensions and draw by using corresponding commands.
5. After that project its top view and repeat the drawing by considering respective draw commands.
6. Similarly complete its side view.
7. Give the required dimensions.
8. Select the location where you want to save the file and click on 'save'
9. Save the document in pdf format.

Solution:



Precautions:

1. Set the drawing screen size to 420 mm × 297 mm. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Follow metric system such that all the dimensions are in mm.

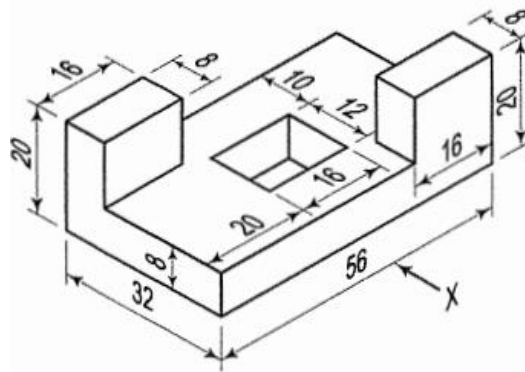
Result: Hence the isometric view of the given orthographic projection of the object is drawn.

Viva Voce:

1. Which function key is used to make OSNAP on/off? []
 - a) F5
 - b) F3
 - c) F4
 - d) F7
2. To dimension the true length of arc _____
 - a) click on the arc, then the two end points
 - b) click on the two end points, then the arc
 - c) click on the arc only
 - d) click on the end points only
3. The method in which the object is placed in the third quadrant is known as _____ method.
 - a) third angle
 - b) second angle
 - c) first angle
 - d) fourth angle
4. In _____ method the plane of projection is transparent.
 - a) third angle
 - b) second angle
 - c) first angle
 - d) fourth angle

Practice No – 12b

Aim: To draw the orthographic projection of the given isometric view of the object.

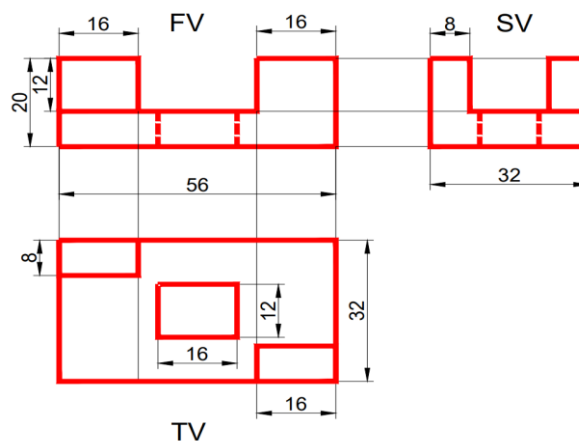


Software Required: AutoCAD 2023

Procedure:

1. Make necessary steps like units, limits and zoom. Pick the ortho mode on.
2. Access the Snap and Grid tab of the Drafting Settings dialog box
3. Observe the given isometric view. The arrow mark indicates the FV direction.
4. In the front view observe dimensions and draw by using corresponding commands.
5. After that project its top view and repeat the drawing by considering respective draw commands.
6. Similarly complete its side view.
7. Give the required dimensions.
8. Select the location where you want to save the file and click on 'save'
9. Save the document in pdf format.

Solution :



Precautions:

1. Set the drawing screen size to 420 mm × 297 mm. ISO A3 dimensions.
2. Follow standards and conventions of drawing properly.
3. Follow metric system such that all the dimensions are in mm.

Result: Hence the isometric view of the given orthographic projection of the object is drawn.

Viva Voce:

1. Orthographic projection is drawn using two methods which is _____
 - a) Second angle and third angle method
 - b) First angle and third angle method
 - c) First angle and fourth angle method
 - d) Second angle and fourth angle method
2. The method in which the object is placed in the first quadrant is known as _____ method.
 - a) third angle
 - b) second angle
 - c) first angle
 - d) fourth angle
3. In first angle method the top view is drawn _____ of the front view.
 - a) above
 - b) right Side
 - c) left side
 - d) bottom
4. In third angle method the top view is drawn _____ of the front view.
 - a) above
 - b) right Side
 - c) left side
 - d) bottom